Football and Racism: Approach Write Up

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November 2023

1 Introduction

Our project leverages Natural Language Processing (NLP), a subfield of cognitive science and Artificial Intelligence, to analyze interactions between computers and human language in the context of football-related discourse. We aim to employ NLP to understand and analyze natural language data extracted from various digital sources.

2 Data Collection Process

1. **Reddit Data Collection**:
   - **API Key Setup**: Follow the guide on obtaining a Reddit API key.
   - **Data Extraction with PRAW**: Utilize PRAW for a more streamlined experience in extracting data from Reddit, including posts and comments from relevant football subreddits.
2. **Instagram Data Collection**:
   - **Instagram API**: Use Instagram's API to collect comments and posts. Given Instagram's API limitations, this may require additional tools or manual methods for more extensive data collection.
   - **Third-Party Tools**: Consider using third-party scraping tools designed for Instagram to gather data from posts, comments, and hashtags related to racism in football.
3. **Collecting from Articles**:
   - **Web Scraping**: Use web scraping tools to collect articles from online newspapers, journals, and blogs. Focus solely on the topics addressing racism in football.
4. **YouTube Data Collection**:
   - **Manual Downloading**: For a smaller dataset, manually download comments directly from YouTube. This method is quick and requires minimal setup.

- **Programmatic Collection**: For larger datasets, use YouTube's API and tools like youtube-dl. This approach is more efficient for collecting comments from a large number of videos but requires a Google account and an enabled API with the **client_secret.json** file.

We are using a range of internet articles and social media sources in order to get a proper representation of the football community on the side of team, management and fans, making sure to include each of their its sentiments. Using APIs are key for extraction of large amounts of data from social media and we want to use them in combination with potential third party scraping tools in scenarios where the API is not giving us the depth of information we require, or where the third party scraping tools give us more relevant data for our project compared to the API. We are using this API/Scraping approach because we are predicting having to handle large amounts of data for each category.

3 Dataset Preparation and Annotation:

For the development and learning of our implicit racism classifier, annotated data is crucial. We will execute a data annotation process where Modal and Meenakshi will review and label datasets for instances of implicit racist language, as well as non racist and explicitly racist instances. This annotated data will serve as the training ground for our classifier, ensuring it learns to detect subtle racist nuances. We will ensure the quality of annotations through reliability checks and iterative review sessions between ourselves. This process will establish a solid foundation for our classifier to learn from accurately labeled data examples which is essential for the nuanced task of identifying implicit racism.

4 Text Preprocessing for NLP Modeling

1. **Tokenization**:
   - The process of splitting text into sentences and words.
   - This stage also involves lowercasing words and removing punctuation to standardize the input for further processing.
2. **Stemming**:
   - A method for reducing words to their base or root form.
   - It typically involves chopping off the ends of words, although this can lead to inaccuracies due to its simplistic approach to the textual data.
3. **Lemmatization**:

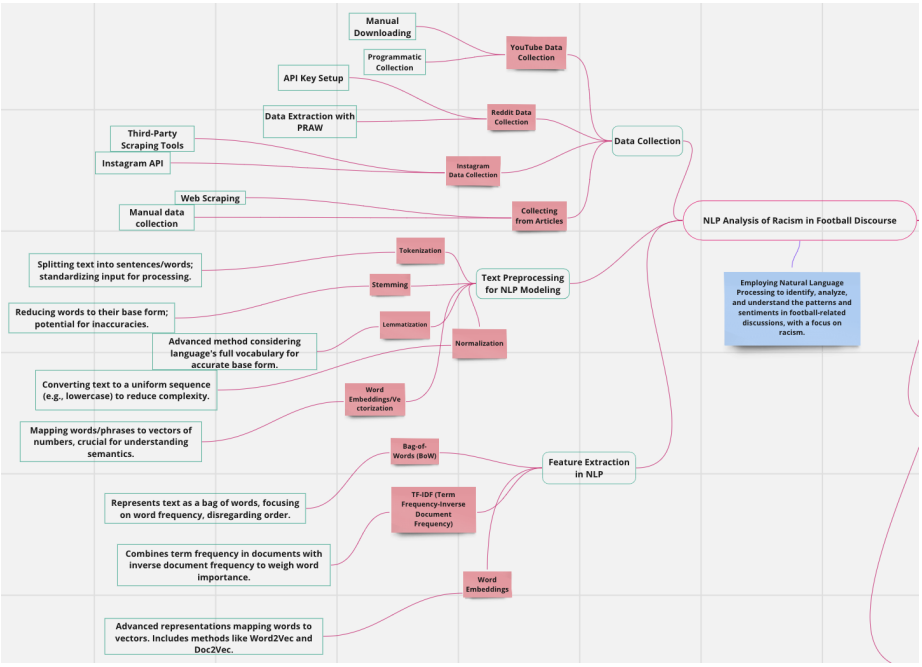
- A more sophisticated method than stemming, lemmatization considers the full vocabulary of a language to apply a morphological analysis to words.
- It involves converting words to their base or dictionary form (e.g., changing verbs in past tense to present).

4. **Normalization**:
   - This process transforms all words in a text to a uniform sequence, such as converting all characters to lowercase.
   - Normalization aids in reducing complexity and variability in the text data.
5. **Word Embeddings/Vectorization**:
   - Involves mapping words or phrases to vectors of real numbers.
   - This process is crucial for enabling algorithms to perform operations like predicting word similarities and understanding semantics.

   We chose to execute one of the standard NLP preprocessing approaches to our dataset to give us a great base for our coming models and NLP analyses.

5 Feature Extraction in NLP

1. **Bag-of-Words (BoW)**:
   - Represents text as a bag (multiset) of words, disregarding grammar and word order but keeping word frequency.
   - Useful in document classification where the occurrence of each word is a feature.
   - The main advantage is its simplicity and effectiveness for certain tasks like language modeling.
2. **TF-IDF (Term Frequency-Inverse Document Frequency)**:
   - **TF (Term Frequency)**: Measures how frequently a term occurs in a document, normalized by dividing by the document length.
   - **IDF (Inverse Document Frequency)**: Assesses the importance of a term, down-weighting terms that appear frequently across documents.
   - **TF-IDF** is a statistical measure that evaluates the importance of a word to a document in a collection or corpus.
3. **Word Embeddings:**

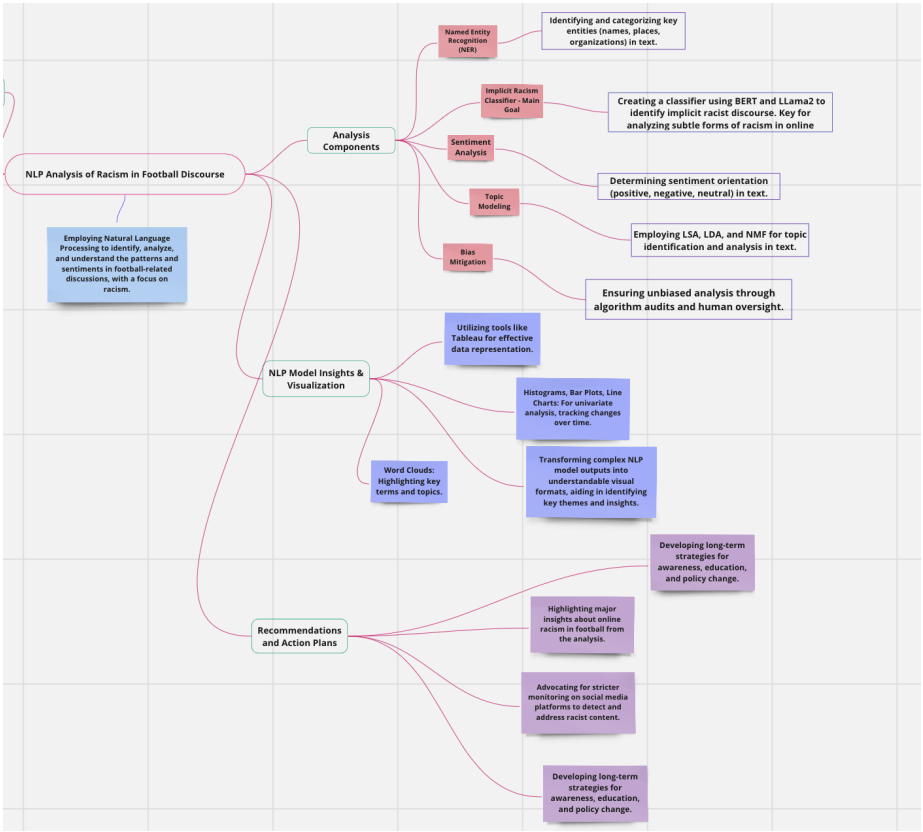


- **Word2Vec**: A method for learning vector representations of words, capturing semantic meanings.
- **Doc2Vec**: Similar to Word2Vec but analyzes groups of text, like entire documents or pages.

6 Data-Driven Roadmap for Racism Analysis in Football Discourse:

7 Main Goal:

Our project's main goal is to create a classifier that can accurately identify and categorize implicit racist discourse within football-related social media content. By fine-tuning advanced Large Language Models such as BERT and LLama2, we will train our system using a rigorously annotated dataset that captures the subtleties of implicit racism. This tool will be at the forefront of our analysis, working in tandem with sentiment analysis, NER, and topic modeling to provide a comprehensive understanding of both overt and subtle forms of racism. Through iterative evaluation and refinement, our classifier aims to be a significant asset in the fight against racism in online football communities.


NLP Analysis of Racism in Football Discourse
Employing Natural Language Processing to identify, analyze, and understand the patterns and sentiments in football-related discussions, with a focus on racism.
Analysis Components
Named Entity Recognition (NER)
Identifying and categorizing key entities (names, places, organizations) in text.
Implicit Racism Classifier - Main Goal
Creating a classifier using BERT and LLama2 to identify implicit racist discourse. Key for analyzing subtle forms of racism in online
Sentiment Analysis
Determining sentiment orientation (positive, negative, neutral) in text.
Topic Modeling
Employing LSA, LDA, and NMF for topic identification and analysis in text.
Bias Mitigation
Ensuring unbiased analysis through algorithm audits and human oversight.
NLP Model Insights & Visualization
Utilizing tools like Tableau for effective data representation.
Histograms, Bar Plots, Line Charts: For univariate analysis, tracking changes over time.
Word Clouds: Highlighting key terms and topics.
Transforming complex NLP model outputs into understandable visual formats, aiding in identifying key themes and insights.
Recommendations and Action Plans
Developing long-term strategies for awareness, education, and policy change.
Highlighting major insights about online racism in football from the analysis.
Advocating for stricter monitoring on social media platforms to detect and address racist content.
Developing long-term strategies for awareness, education, and policy change.

8 Integrating Text Classification and Topic Modeling in Racism Analysis for Football Discourse:

Objective: The primary goal is to utilize topic modeling and text analysis to systematically rank and categorize hosting countries, regional fan bases, or football leagues based on the prevalence and nature of racist discourse. By analyzing vast amounts of textual social media data, we aim to identify and quantify key themes and patterns of racism. This will provide a data-driven assessment of racism's prominence in the football world.

Additionally, our approach will:

- Track Sentiment Trends: Analyze how sentiments about racism evolve over time or in response to key events.
- Conduct Geographical Analysis: Examine variations in discussions about racism across different countries or regions.
- Identify Key Influencers: Pinpoint influencers or opinion leaders in the discussions on racism in football.
- Perform Predictive Analysis: Use historical data to anticipate future trends or potential incidents.
- Analyze Behavioral Patterns: Understand the behavior of users engaging in racist discourse.

Text Classification: We will employ both rule-based and machine learning systems for text classification. Rule-based systems will flag specific keywords related to racism or football, and machine learning models will learn from historical data to identify patterns of racism.

Feature Extraction: Previously explained techniques like Bag-of-Words and TF-IDF will be utilized.

Algorithms: We'll use algorithms such as Decision Trees, Naive Bayes, Support Vector Machines, and Neural Networks for classification. Evaluation: Performance will be evaluated using accuracy, precision, recall, and F1-score.

Topic Modeling: We are implementing Latent Semantic Analysis (LSA) and Latent Dirichlet Allocation (LDA) to dissect and understand the massive volume of social media data concerning racism in football.

Efficiency with Short Texts: We are specifically applying Non-Negative Matrix Factorization (NMF) to efficiently handle the brief and often informal text that characterizes social media content.

Data Interpretability: To improve the clarity and interpretability of our data, we're exploring dimensionality reduction techniques like Random Projection and Principal Component Analysis (PCA).

Evaluation Metrics: We will utilize standard statistical metrics, including recall, precision, F-score, and topic coherence, to evaluate the efficacy of our topic modeling methods.

9 Sentiment Analysis

Sentiment analysis will be employed to determine the sentiment orientation (positive, neutral, negative, and more granular emotions like happiness, sadness, frustration) in text related to racism in football. This analysis will sift through social media, forums, and other digital platforms to gauge public sentiment about racism in various football contexts. By employing machine learning techniques, we aim to process vast amounts of data, overcoming the "noise" inherent in large datasets. The outcome will be a nuanced understanding of public sentiment towards racism in football, supported by data visualizations that highlight sentiment trends across different regions, fan bases, and events.

10 Named Entity Recognition

In our project, N.E.R will be used to automatically identify and categorize key information from unstructured text related to racism in football. This includes recognizing names of players, teams, organizations, locations, and specific events. The application of N.E.R will help in pinpointing specific entities involved in racist incidents or discussions, facilitating a more targeted analysis. For example, N.E.R can identify which players, teams, or matches are most frequently mentioned in the context of racism, thereby providing deeper insights into the specific areas of concern within the football community.

11 Bias Mitigation

: Our objective is to ensure fair, accurate, and unbiased analysis of racism in football, especially when using Large Language Models (LLMs). We plan to use LLMs trained on diverse datasets to minimize cultural and contextual biases. Regular algorithmic audits will be conducted to check for any biases that could affect sentiment analysis or NER results. Human oversight will play a crucial role in reviewing and validating findings, particularly in ambiguous cases. Continuous monitoring of model performance will help in identifying emerging biases. Additionally, we will collaborate with linguists, sociologists, and domain experts to further understand and mitigate potential biases in our analysis.

12 NLP Model Insights & Visualizations:

Data Representation: Utilize visualization tools like Tableau to represent both unstructured text data and structured outputs from our NLP models. This includes visualizing word frequencies, sentiment trends, and named entity occurrences.

Visualization Techniques:

- Word Clouds: Display key terms and topics identified in the discourse, highlighting the most frequent and significant words.
- Histograms, Bar Plots, and Line Charts: Use these for univariate analysis, such as tracking sentiment changes over time or comparing the frequency of topics across different regions or fan bases.
- Interpreting Model Outputs: Transform complex NLP model outputs into understandable visual formats to identify key themes and insights, such as the prevalence of certain sentiments or the association of specific entities with racism.
- Analyzing and Reporting: Provide comprehensive reports based on visualizations to communicate findings effectively to stakeholders, offering clear and actionable insights.

13 Recommendations and Action Plans

Our findings highlight the urgent need for stronger measures against online racism in football. Despite efforts by Twitter and Facebook to remove abusive content, there have been instances where racist comments remained visible for hours or were not removed even after being reported. The Football Association's statement condemning the racism faced by England players on social media underscores this issue. Major English football bodies have criticized these platforms for failing to prevent the spread of racism and harassment. The Premier League's boycott of social media platforms in April to demand change and eradicate online hate further emphasizes the need for social media companies to intensify their efforts. Our recommendations include advocating for stricter online monitoring, prompt action against racist content, and collaborative efforts with governing bodies and leagues to create a more accountable online environment.

14 Conclusion

In summary, our project leverages NLP techniques to analyze and understand the discourse around racism in football on digital platforms.

Through sentiment analysis, topic modeling, and named entity recognition, we aim to uncover and quantify prevalent themes, gauge public sentiment, and identify key entities involved in racist discourse. The project emphasizes the necessity of bias mitigation and effective data visualization to ensure accurate and comprehensible insights. Our recommendations focus on proactive measures against online racism and call for greater accountability from social media platforms. This research not only contributes to understanding racism in football but also advocates for tangible changes in online and offline environments.